**Day 19 — Broadcast Join : Scenario: Join millions of transactions with a small branch master.**

**COMMANDS :**

Create project

```
mkdir -p ~/day19-broadcast-join/src/main/scala
cd ~/day19-broadcast-join

nano build.sbt

ThisBuild / version := "0.1.0"
ThisBuild / scalaVersion := "2.12.18"

lazy val root = (project in file("."))
  .settings(
    name := "Day19-Broadcast-Join",

    libraryDependencies ++= Seq(
      "org.apache.spark" %% "spark-core" % "3.5.3",
      "org.apache.spark" %% "spark-sql" % "3.5.3"
    ),

    Compile / run / fork := true,

    Compile / run / javaOptions ++= Seq(
      "--add-exports=java.base/sun.nio.ch=ALL-UNNAMED"
    )
  )

nano src/main/scala/BroadcastJoin.scala

import org.apache.spark.sql.SparkSession
import org.apache.spark.sql.functions.broadcast
object BroadcastJoin {
  def main(args: Array[String]): Unit = {
    val spark = SparkSession.builder()
      .appName("Day19 Broadcast Join")
      .master("local[*]")
      .getOrCreate()
    spark.sparkContext.setLogLevel("WARN")
    import spark.implicits._
    val transactions = Seq(
```

```
  (1, 101, 1001, 5000.0),
  (2, 102, 1002, 2500.0),
  (3, 101, 1003, 1500.0),
  (4, 103, 1004, 8000.0),
  (5, 104, 1005, 3000.0),
  (6, 102, 1006, 4500.0),
  (7, 105, 1007, 6000.0)
)
val transactionDF = transactions.toDF(
  "transaction_id",
  "branch_id",
  "customer_id",
  "amount"
)
val branches = Seq(
  (101, "Main Branch", "Hyderabad", "Telangana"),
  (102, "Central Branch", "Vijayawada", "Andhra Pradesh"),
  (103, "North Branch", "Warangal", "Telangana"),
  (104, "South Branch", "Chennai", "Tamil Nadu"),
  (105, "East Branch", "Visakhapatnam", "Andhra Pradesh")
)
val branchDF = branches.toDF(
  "branch_id",
  "branch_name",
  "city",
  "state"
)
println("\n========== TRANSACTIONS ==========")
transactionDF.show()
println("\n========== BRANCH MASTER ==========")
branchDF.show()
val resultDF = transactionDF
  .join(
    broadcast(branchDF),
    transactionDF("branch_id") === branchDF("branch_id"),
    "inner"
  )
  .select(
    transactionDF("transaction_id"),
    transactionDF("customer_id"),
    transactionDF("branch_id"),
    branchDF("branch_name"),
    branchDF("city"),
    branchDF("state"),
```

```
      transactionDF("amount")
    )
  println("\n========== BROADCAST JOIN RESULT ==========")
  resultDF.show(false)
  println("\n========== EXECUTION PLAN ==========")
  resultDF.explain()
  spark.stop()
 }
}
```

```
sbt clean
sbt compile
sbt run
```

**OUTPUTS :**

```
========== TRANSACTIONS ==========
+--------------+---------+-----------+------+
|transaction_id|branch_id|customer_id|amount|
+--------------+---------+-----------+------+
|             1|      101|       1001|5000.0|
|             2|      102|       1002|2500.0|
|             3|      101|       1003|1500.0|
|             4|      103|       1004|8000.0|
|             5|      104|       1005|3000.0|
|             6|      102|       1006|4500.0|
|             7|      105|       1007|6000.0|
+--------------+---------+-----------+------+

========== BRANCH MASTER ==========
+---------+--------------+-------------+--------------+
|branch_id|   branch_name|         city|         state|
+---------+--------------+-------------+--------------+
|      101|   Main Branch|    Hyderabad|     Telangana|
|      102|Central Branch|   Vijayawada|Andhra Pradesh|
|      103|  North Branch|     Warangal|     Telangana|
|      104|  South Branch|      Chennai|    Tamil Nadu|
|      105|   East Branch|Visakhapatnam|Andhra Pradesh|
+---------+--------------+-------------+--------------+

========== BROADCAST JOIN RESULT ==========
+--------------+-----------+---------+--------------+-------------+--------------+------+
|transaction_id|customer_id|branch_id|branch_name   |city         |state         |amount|
+--------------+-----------+---------+--------------+-------------+--------------+------+
|1             |1001       |101      |Main Branch   |Hyderabad    |Telangana     |5000.0|
|2             |1002       |102      |Central Branch|Vijayawada   |Andhra Pradesh|2500.0|
|3             |1003       |101      |Main Branch   |Hyderabad    |Telangana     |1500.0|
|4             |1004       |103      |North Branch  |Warangal     |Telangana     |8000.0|
|5             |1005       |104      |South Branch  |Chennai      |Tamil Nadu    |3000.0|
|6             |1006       |102      |Central Branch|Vijayawada   |Andhra Pradesh|4500.0|
|7             |1007       |105      |East Branch   |Visakhapatnam|Andhra Pradesh|6000.0|
+--------------+-----------+---------+--------------+-------------+--------------+------+
```